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Remotely controlled smart monitoring system of hermetic paddy storage to reduce postharvest losses in Bangladesh

Establishes scientific basis for storage parameters

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ABSTRACT

In Bangladesh, where agriculture forms the backbone of the economy, efficient grain storage is crucial, particularly for paddy storage. Traditional methods like gunny bags and plastic containers are common, but the recent introduction of hermetic storage technologies, such as GrainPro bags and hermetic cocoons, have significantly reduced postharvest losses paddy. However, most research focuses on oxygen (O_2) and carbon dioxide (CO_2) levels in those open storage systems. This study introduces an innovative, cost-effective remote monitoring system for hermetically sealed grain storage, designed using locally sourced components, including a Digital Hygrometer, WiFi Mini IP Camera, and a Remote Controller unit. Tested over a 125-day period at Advanced Storage Lab, Bangladesh Agricultural University, the system effectively monitored the internal environment of the storage units. Key findings show that the internal temperature of the storage bags varied between 16.7°C and 29.2°C, demonstrating lesser fluctuation compared to the ambient temperature range of 16.1°C to 33.5°C. The relative humidity inside the storage units remained stable between 45 % and 54 %, in contrast to the external humidity range of 45 % to 81.2 %. Additionally, using the Steffe and Singh equation, an increase of 0.512 % in Equilibrium Moisture Content (EMC) of paddy was recorded. Notably, there was no evidence of insect or mold growth after 125 days of storage. This study's remote monitoring system not only marks a significant advancement in hermetic grain storage technology but also contributes to sustainable agricultural practices. It provides a practical, real-time solution to monitor and manage key environmental parameters, ensuring the preservation of grain quality over time.

1. Introduction

Grain storage plays a pivotal role in agricultural sustainability, with significant implications in rice-dependent economies like Bangladesh. The integrity of stored grain, profoundly influenced by factors such as moisture content, temperature, and environmental conditions, determines the quality and nutritional value of the grain. These factors are crucial in mitigating grain degradation, insect infestation, and mold growth, making the balance of moisture and temperature vital [1,2]. Direct measurement of temperature (T) and relative humidity (RH) within grain masses is essential. This practice aids in assessing grain quality degradation and determining allowable storage durations. Understanding Equilibrium Moisture Content (EMC), calculated from T and RH, is key to gauging the condition of stored grains and ensuring their high quality [3]. In Bangladesh, post-harvest losses, particularly for paddy, are alarming, estimated at around 14 %. These losses are largely due to inadequate drying and improper storage methods. Traditional storage methods, such as gunny and plastic bags, fall short in effectively controlling critical factors like humidity and temperature. Conversely, innovative solutions like GrainPro bags and Hermetic Cocoons have emerged, offering more controlled environments and enhancing grain preservation [4,5].

Hermetic storage bags, effective against fungal spoilage, insect infestation and mycotoxin accumulation, ensure the safety of stored produce [6]. The impact of storage conditions on grain quality is further exemplified in studies by A. A. Sawant and Strelec, revealing how environmental factors influence moisture content and germination rates [7,8]. Similarly, recent advancements in smart monitoring systems, developed by teams like Michael A. Omodara's, highlight the potential of technology in revolutionizing grain storage monitoring. With the rise in global grain production, efficient warehousing becomes increasingly critical for market success and profitability [9]. This review, drawing on

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existing storage systems, care procedures, and pivotal studies, underscores the necessity of maintaining grain quality to meet consumer demands [10]. Focusing on the critical role of humidity in pest control within hermetic paddy storage, studies using GrainPro bag have assessed varying humidity levels and temperatures, illuminating the interplay between these factors in controlling pests and preserving paddy quality [11]. The advent of IoT in grain storage, facilitating real-time monitoring through wireless sensor networks, represents a significant leap in storage efficiency and quality control [12,13]. Comparative research, like that conducted by Gervancio Covele, has shown the superiority of hermetic storage containers over traditional methods, significantly reducing infestation levels and grain damage. This underscores the environmental benefits of such technologies for smallholder farmers in food security and income generation [14]. Non-residue organic technologies like Modified Atmospheres (MAs), responding to pesticide restrictions, are gaining traction for protecting stored grain, satisfying consumer preferences for chemical-free products [15]. Hermetic storage technology has evolved globally, with innovations like remote monitoring systems and the Cocoon Lite™, enhancing its effectiveness in diverse climatic conditions [16]. Studies comparing various storage bags, such as those by D.O. Kiobia, highlight the effectiveness of specific types like Purdue Improved Crop Storage (PICS) bags in maintaining low humidity and minimizing insect damage [17–19]. The introduction of automated grain monitoring solutions like the HDS Tara system, employing IoT communication technology, exemplifies the move towards modern, cost-effective storage methods suitable for small to medium-sized landowners [20,21]. Innovative humidity monitoring solutions in hermetically packaged food commodities, utilizing advanced sensing technologies, have been developed to further optimize storage conditions [16]. In-depth studies on the impact of storage temperature and relative humidity on various grains, including those by Jon González-Torralba and Chan-Eun Park, contribute to our understanding of optimal storage conditions [22,23]. These studies highlight the intricate balance required to preserve grain quality, emphasizing the potential benefits of advanced technologies in extending storage life and reducing losses due to factors like discoloration [24].

Mihajlo et al. (2020) introduced a novel grain monitoring system leveraging NB-IoT technology. Their setup incorporated HYGRO boards equipped with HDC1080 sensors housed in PVC probes. Despite its ingenuity, their chosen sensor proved inadequate for precise moisture content monitoring of grains. Gargelwar and Kuntawar (2019) proposed another grain storage monitoring solution, featuring an Arduino Nano controller, NRF24L01 transceivers, and DHT11 sensors for temperature and relative humidity measurements. While innovative, these systems lacked the precision necessary for comprehensive grain monitoring. In a similar vein, Lutz and Coradi (2023) engineered a real-time Equilibrium Moisture Content and Carbon Dioxide monitoring system to forecast corn grain quality stored in silo bags, employing ESP8266 D1 mini, DHT22/AM2302, and MHZ-19 components, augmented by Artificial Neural Networks. Although promising, the complexity and costliness of these systems pose notable challenges.

Despite these advancements, challenges in grain storage persist, especially in Bangladesh. Current monitoring systems often face issues like high costs, delayed data, and inadequate airtight conditions. This study aims to address these challenges by developing a Remote-Control Monitoring System for Hermetic Paddy Storage. Leveraging cutting-edge technology, this research strives to offer a cost-effective, real-time solution to enhance grain storage monitoring, thereby aiming to reduce post-harvest losses and bolster food security in the region.

2. Material and methods

2.1. Advanced monitoring system design

The cornerstone of the research involved the strategic implementation of a wireless, integrated monitoring system, meticulously engineered to observe and record environmental conditions within a hermetically sealed GrainPro bag used for paddy storage. This system represents a confluence of technology and practical design, aimed at enhancing the accuracy and efficiency of grain storage monitoring.

2.1.1. Monitoring system preparation

The development of the monitoring system began with the procurement of materials from local markets. The centerpiece, a digital hygrometer, underwent calibration in the Advanced Storage Lab at Bangladesh Agricultural University (BAU). This calibration process was critical, aligning the hygrometer with lab standard hygrometer (HT-3007SD) and achieving an average temperature error of 1.09 % and relative humidity error of 0.99 %.

Subsequently, the assembly involved meticulous placement and securing of the camera, remote control unit, and battery set at the base of a customized box using super glue for stability. Surrounding the camera, two LED lights were installed to ensure adequate illumination. These components, including the camera, remote control unit, battery, and LED lights, were interconnected through precise soldering, forming an integrated circuit. Fusion 360 software was employed to engineer sophisticated 3D visualization, enhancing the simulation of the system for comprehensive understanding and facilitating advancements as shown in Fig. 1. Positioned strategically atop the enclosure, directly opposite the camera, is a digital hygrometer securely affixed using a robust adhesive like super glue. To optimize accuracy in environmental readings, it was intricately designed a perforated compartment on one side of the box, exclusively housing the sensor of the digital hygrometer. This meticulous arrangement ensures precise monitoring of the storage environment. Finally, the entire system was meticulously sealed to safeguard its components and uphold the integrity of its functionality.

2.1.2. Component overview and operational dynamics

The heart of the experimental monitoring framework is a sophisticated digital hygrometer, capable of measuring a wide range of temperatures and relative humidity. This instrument's precision is pivotal for the study's accuracy.

Hygrometer integration for accurate environmental assessment

In the heart of the experimental framework lies an ultra-modern digital hygrometer, a cornerstone in the precise evaluation of critical environmental variables. This cutting-edge instrument is equipped with sophisticated sensor technology, adept at discerning minute temperature changes ranging from the frigid −50ºC to a balmy +70ºC. It's equally proficient in detecting a wide spectrum of relative humidity, spanning from a dry 10 % RH to a highly moist 99 % RH. Such a range ensures comprehensive coverage of environmental conditions, pivotal

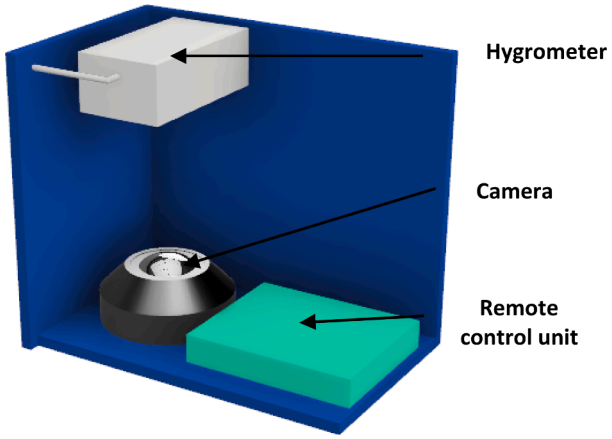


Fig. 1. 3D modelling of the compact system.

for the study's accuracy.

Power efficiency and longevity

To facilitate uninterrupted operation over substantial durations, the hygrometer is energized by a pair of LR44 button batteries. These batteries are not just ordinary power sources; they are engineered to deliver consistent energy output, boasting an impressive lifespan of up to six months under regular usage. This longevity is crucial in minimizing maintenance intervals and ensuring continuous data collection.

Calibration and validation for unmatched precision

Underpinning the reliability of this hygrometer is a thorough calibration process, conducted in the state-of-the-art Advanced Precision Lab within the Department of Farm Power and Machinery (FPM) at BAU. This process involved a meticulous comparison against a standard hygrometer (HT-3007SD), renowned for its accuracy. The outcome of this rigorous calibration was remarkable – a mere 1.09 % deviation in temperature readings and an even more impressive 0.99 % variance in relative humidity measurements. Such precision is not just commendable; it's critical in reinforcing the credibility of environmental monitoring endeavors.

Compact real-time data transmission camera

The experiment employs a compact yet powerful real-time data transmission camera (A9 WiFi Mini IP), essential for capturing and transmitting real-time data. This camera, notable for its capability to record continuously for up to 30 min within a range of 3 m, is pivotal in documenting the environmental sensor's readings. It is equipped with enhanced battery features to ensure consistent performance throughout the duration of the study.

Radio frequency-based remote controller unit

The remote controller unit, functioning on radio frequency signals, is a cornerstone of the system, facilitating the seamless operation of both the camera and an auxiliary LED light to ensure optimal visibility in varying light conditions. This unit's operational efficacy extends over a range of 10 to 20 m, ideal for open-area applications.

2.2. GrainPro bag and paddy as the experimental subject

The experimental study was conducted using a GrainPro bag, renowned for its hermetic sealing capabilities, as the storage medium. Filled with the locally significant *BRRI dhan49* paddy variety, the GrainPro bag not only mirrored the regional agricultural context but also served as a robust test medium for evaluating the efficacy of the monitoring system. This setup provided an ideal scenario to assess the system's performance in real-world conditions, ensuring the relevance and applicability of the research findings. Through this detailed overview of the system components and their functionalities, it can demonstrate the thoughtfulness and innovation embedded in the design of the monitoring system. Each component's role is critical to achieving the overarching goal of providing real-time, accurate environmental monitoring within grain storage facilities, thereby contributing to improved storage management and grain quality preservation.

2.2.1. Grain preparation for storage

For the study, 40 kg of freshly harvested *BRRI dhan49* paddy was procured from the local market, initially with a moisture content of 14.3 %. This paddy was then subjected to a three-day sun-drying process to reduce the moisture level to 10.83 %. Post-drying, the paddy was meticulously cleaned to remove any unwanted materials, preparing it for storage.

2.2.2. Monitoring system implementation

A GrainPro bag was chosen for storage due to its hermetic sealing capabilities. The prepared paddy was filled half of the bag. The monitoring system was then strategically placed in the middle section of the bag, ensuring optimal positioning for accurate data collection. Following this, the remainder of the paddy was added, and the bag was tightly sealed. This implementation process ensured that the monitoring system was centrally located within the grain mass, allowing for comprehensive environmental data collection throughout the storage period. The sealed GrainPro bag, with its contents and embedded monitoring system, represented a microcosm of typical storage conditions, enabling to assess the system's efficacy under realistic scenarios.

2.2.3. Deployment strategy

The deployment strategy for real-time environmental monitoring in grain storage facilities begins with the activation of a command device. This device, designed for user ease, sends a continuous radio signal to a specialized monitoring box placed within a GrainPro bag filled with paddy as shown in Fig. 2. The monitoring box is equipped with a fitted remote-control unit that is highly sensitive to these signals. Upon receiving the signal, the remote-control unit activates two critical components simultaneously: a camera and an LED light. This synchronization is achieved through an efficient electrical circuit that draws power from an onboard battery, ensuring a consistent energy supply. The camera, once powered, establishes a wireless connection with a cell phone via Wi-Fi. This step is crucial for real-time data transmission. When the connection is established (Fig. 3), the camera relays crucial environmental data specifically, the temperature and relative humidity inside the paddy storage directly to the cell phone display. This real-time data acquisition is pivotal for monitoring the microenvironment within the grain storage, which is essential for maintaining grain quality and preventing spoilage. For data logging and analysis, two approaches are suggested: taking a screenshot of the data displayed on the cell phone or manually recording the readings on a data sheet. This dual approach to data collection allows for flexibility in data analysis and record-keeping. The system is designed to cease operation upon pressing the 'off' button on the command device. This action cuts off the radio signal to the box, resulting in the deactivation of both the camera and the LED light. This feature not only conserves battery life but also ensures that the monitoring is done only when needed, thereby enhancing the efficiency and longevity of the monitoring system.

2.3. Comprehensive data recording and analytical approach

Data collection and statistical analysis

In the context of Bangladesh, where grain storage typically spans approximately 3 months (120 days), this study extended over a period of 125 days. The research employed a meticulously structured data collection process utilizing an integrated monitoring system housed within a GrainPro bag. Data collection involved intensive daily observations method within the GrainPro bag, alongside concurrent recordings of parameters in the room environment for comparative analysis. This dual approach, monitoring both internal conditions and external factors, was essential for thorough analysis. Subsequently, the collected data underwent rigorous analysis to determine correlations and regression patterns, yielding a comprehensive dataset that not only reflected microclimate conditions within the grain storage but also elucidated their interaction with external environmental factors. This methodological approach played a pivotal role in acquiring nuanced insights into the environmental impacts on grain preservation.

2.3.1. Equilibrium moisture content (EMC) estimation

To evaluate the moisture content of the stored paddy, Steffe and Singh equation, a widely acknowledged method in grain moisture content analysis, was employed. The formula used is as follows:

$$EMC = 0.326 - 0.046ln[-1.987 \times (Ti + 35.7) \times ln(RH)] \quad (1)$$

Here, EMC denotes the dry-basis equilibrium moisture content. The equation integrates the recorded temperature, Ti (°C) of interstitial air and the relative humidity, RH (%) of interstitial air, to calculate the EMC [25].

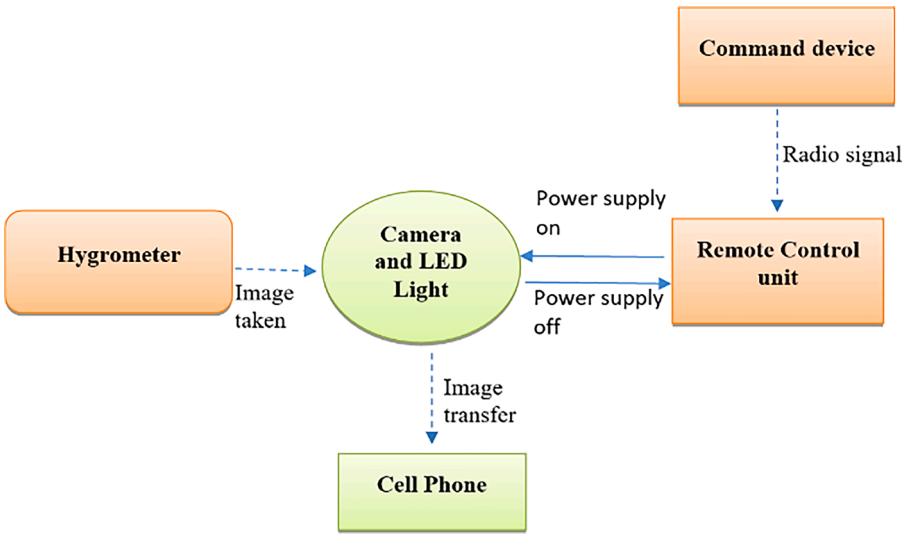


Fig. 2. Entire connection diagram.

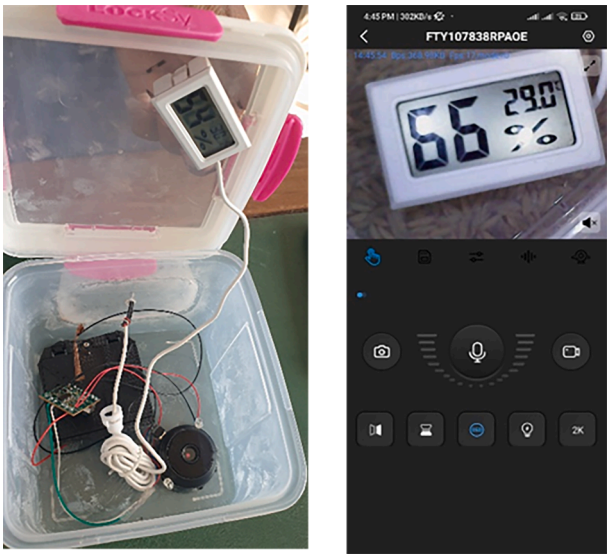


Fig. 3. Working process and monitoring via android apps.

2.3.2. Data interpretation and insights

The application of the Steffe and Singh equation allowed for an accurate assessment of the paddy's moisture content under varying storage conditions. By analyzing the EMC values in conjunction with the recorded temperature and humidity data, valuable insights into the effectiveness of the GrainPro bag in maintaining optimal storage conditions was attained. This analysis was pivotal in understanding the impact of environmental factors on grain quality and longevity.

2.3.3. Implications for grain storage

The comprehensive data analysis provided not only a detailed understanding of the environmental conditions within the hermetic storage but also laid the foundation for developing improved storage strategies. The insights gained from this study are expected to contribute significantly to the field of agricultural storage, particularly in enhancing the methods used for preserving grain quality and preventing post-harvest losses.

2.4. Monitoring system price

The monitoring system was designed with cost-effectiveness in mind, utilizing locally available materials to ensure affordability for Bangladeshi farmers. Table 1 delineates the specifications and prices of the necessary components, revealing a total system cost of a mere 30 USD. This approach underscores a commitment to practicality and accessibility, enabling widespread adoption and empowering farmers with essential monitoring capabilities at a nominal expense.

3. Results

This section delineates the findings from the extensive 125-day study on paddy storage in GrainPro bags, highlighting key environmental parameters: room and inside storage temperatures, relative humidity, and equilibrium moisture content (EMC). These insights are pivotal in evaluating the efficacy of GrainPro bags in preserving grain quality under varying environmental conditions. The data, analyzed through rigorous statistical methods, offers a detailed understanding of the microclimatic dynamics within the storage system and their implications for agricultural storage practices. The results presented here are crucial for advancing grain storage technologies and methodologies.

3.1. Power consumption

The developed remotely controlled smart monitoring unit operates on a DC power source, requiring only a small amount of power for its functionality. Table 2 details the power consumption of each component, with a total daily power consumption of just 3.76 W. This efficient power usage underscores the unit's practicality and sustainability in operation.

Table 1
Price of required items of monitoring system.

Sl. No.	Items	Unit	Price (BDT.)
1.	Web Camera	1	1000
2.	Remote Control Unit	1	600
3.	Box	1	100
4.	GrainPro Bag	1	120
5.	PVC bag	1	80
6.	Paddy	40 kg	1200
		Total (BDT)	3100
		Total (USD)	30

Table 2
Power consumption of the unit.

Item	Power consumption	Operating time/ day	Power consumption/ day
Remote Control unit	*5 mW/sec*	1 min	0.3 W
Wifi camera	*64.86 mW/sec*	40 s	2.6 W
Hygrometer	*0.01 mW/sec*	24 hr	0.86 W
Total power consumption/day			3.76 W

3.2. Temperature monitoring

The study meticulously recorded room and inside storage temperatures over a 125-day period, highlighting the thermal dynamics within the GrainPro bag. The room temperature exhibited significant variability, ranging from 16.1°C to 33.5°C, reflective of the external climatic conditions as shown in Fig. 4. In comparison, the GrainPro bag's internal temperature was relatively stable, ranging from 16.7°C to 29.2°C. This reduced variability points towards the bag's material properties, offering a limited but notable thermal buffer. A detailed thermodynamic analysis suggests that the polyethylene composition of the GrainPro bag, while thin, provides a semi-permeable barrier that moderates the impact of external temperature fluctuations.

3.3. Relative humidity monitoring

The relative humidity (RH) inside the GrainPro bag exhibited a distinct pattern of stability compared to the external environment. The external RH showed (Fig. 5) significant fluctuations between 45 % and 81.2 %, reflecting atmospheric moisture changes. In contrast, the internal RH maintained a narrower band, stabilizing between 45 % and 54 %. This data underscores the bag's efficacy in creating a microenvironment less susceptible to external humidity variations, crucial for grain preservation.

3.4. Equilibrium moisture content (EMC) observations

The Equilibrium Moisture Content (EMC) within the GrainPro bag demonstrated a subtle yet consistent rise from 12.033 % to 12.545 % throughout the storage duration, as depicted in Fig. 6. This incremental shift of 0.512 % underscores the grain's inherent ability to absorb moisture, reflecting the interplay between internal respiration dynamics and external humidity variations. Contrarily, environmental EMC exhibits more frequent fluctuations compared to storage EMC. A comprehensive physicochemical examination elucidates the intricate moisture interaction mechanisms within the storage environment, shedding light on the hygroscopic traits of paddy and their implications for prolonged storage efficacy.

3.5. In-Depth regression analysis

In this comprehensive study, conducted a thorough regression analysis to investigate the changes in Equilibrium Moisture Content (EMC) of paddy during storage. Notably, found that the EMC of paddy increased over the storage period. This trend is attributed to the fact that the internal relative humidity within the GrainPro bag remains largely unaffected by outside environmental changes. This analysis also uncovered a strong Pearson correlation coefficient of 0.875 between the storage period and EMC. This high correlation indicates a significant and direct relationship between the length of storage period and the increase in EMC.

3.6. Model development and implications

3.6.1. Relation between storage EMC and storage time

Building on these findings, a regression model is developed, illustrated in Fig. 7 as EMC (%) versus storage period (Days). The model, boasting an R^2 of 76.8 %, demonstrates a substantial predictive capability for changes in EMC over time. This high degree of predictiveness is a crucial tool in anticipating changes in grain quality during extended storage periods. The regression model's equation enables to estimate the EMC of paddy stored inside the GrainPro bag for any given storage day.

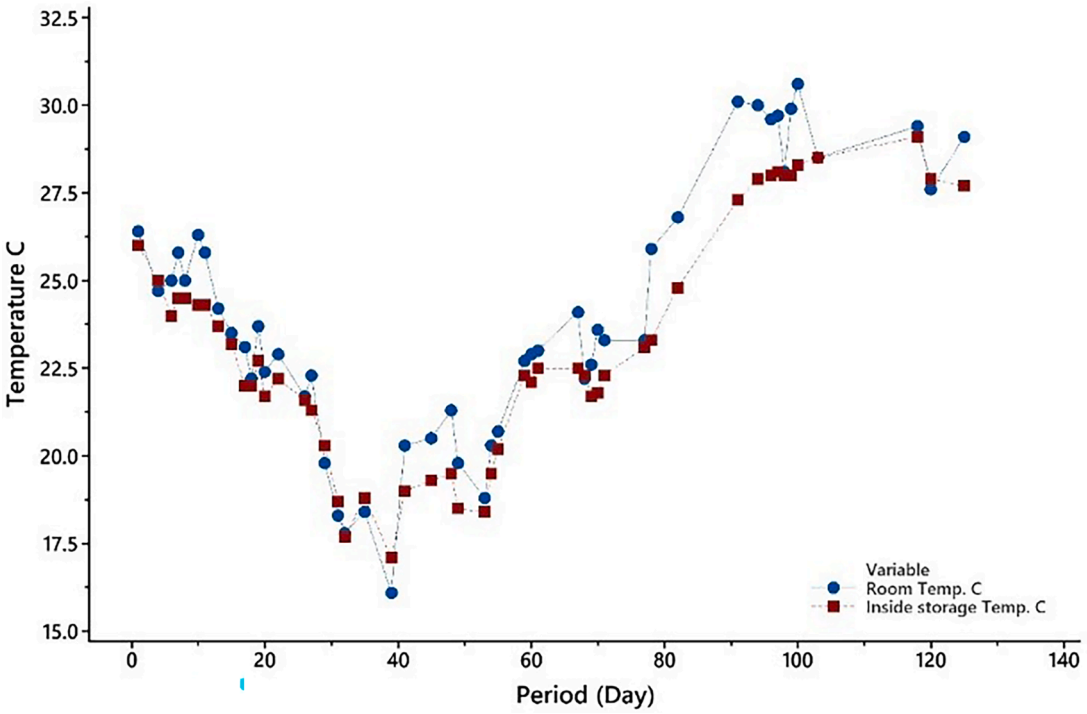


Fig. 4. Temperature data visualization and comparison.

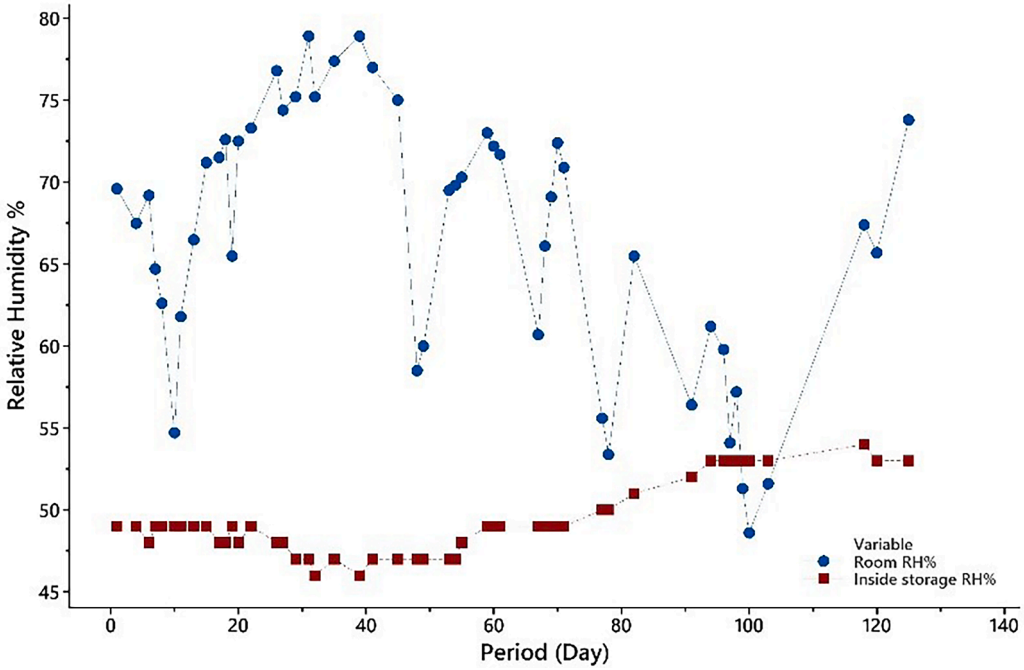


Fig. 5. RH data comparison.

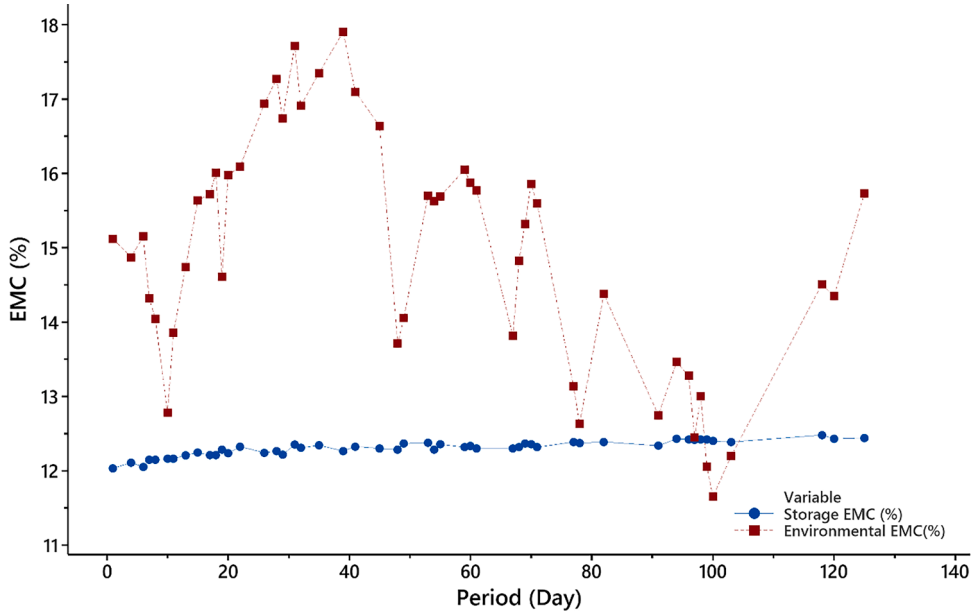


Fig. 6. EMC visualization.

This is particularly beneficial as it allows for assessing the grain condition inside the bag without needing to directly measure temperature and relative humidity. Such a predictive tool is invaluable for grain storage management, enabling better planning and strategies to maintain optimal grain quality and reduce potential losses during storage.

This in-depth regression analysis provides essential insights into the dynamics of paddy storage. The strong correlation between storage period and EMC, coupled with the predictive power this regression model, offers a novel approach to predicting and managing grain quality over time, marking a significant advancement in the field of agricultural storage science.

3.6.2. Relation between storage EMC (%) and storage RH (%)

Fig. 8 illustrates the correlation between inside relative humidity and EMC, revealing a robust relationship. Pearson's correlation coefficient of 0.526 indicates a strong correlation between the two factors, supported by a significant t-value of 4.3 for storage EMC. Furthermore, the influence of inside temperature on storage EMC was found to be insignificant, highlighting the dominance of relative humidity in affecting EMC.

3.6.2. Relation between storage temperature and room temperature

Fig. 9 explains the relationship between the inside temperature of the developed system and the room temperature through a residual plot of inside temperature. This model exhibits an impressive R^2 value of 0.952 and a Pearson's correlation coefficient of 0.975, indicating a strong and

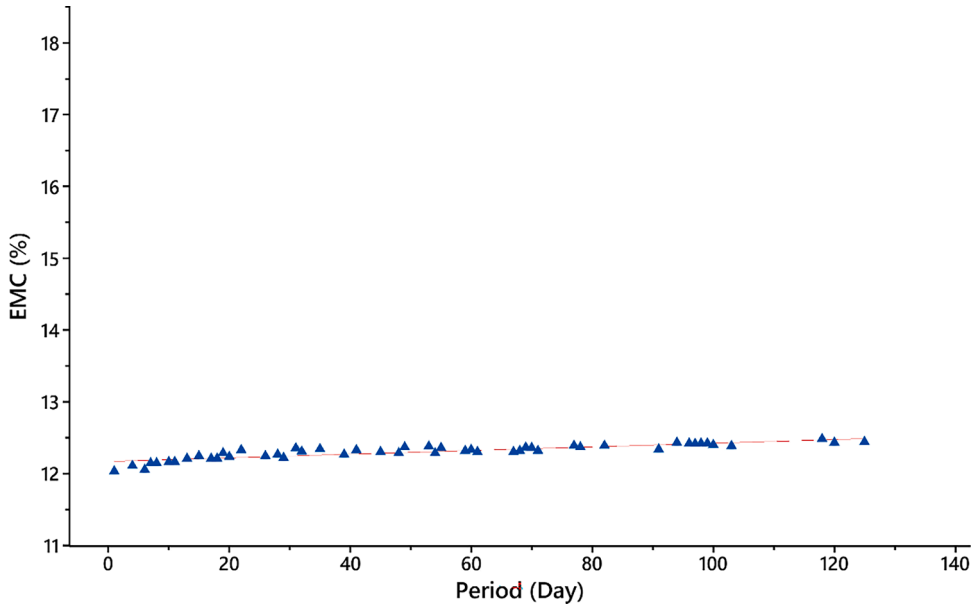


Fig. 7. Regression analysis: EMC (%) versus period (day).

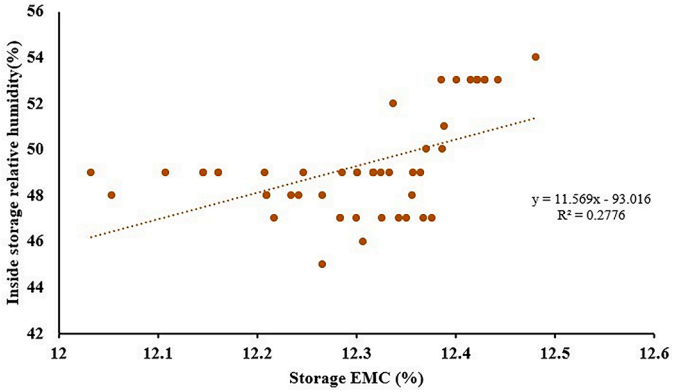


Fig. 8. Regression analysis: EMC (%) versus relative humidity.

positive correlation between the two temperatures. The residual plot underscores the significance of this relationship, demonstrating how the temperature within the storage system is closely tied to the ambient temperature. This regression model provides valuable insights into the dynamics of temperature regulation within the storage environment.

3.7. Comprehensive interpretation and implications

The data reveals a nuanced understanding of the microclimatic conditions within GrainPro bags and their effectiveness in paddy storage. The observed stability in temperature and RH, along with the controlled increase in EMC, highlights the GrainPro bag's suitability for long-term grain storage. This study not only reaffirms the bag's utility in maintaining grain quality but also contributes to the broader understanding of grain storage dynamics under varying environmental conditions. The technical insights provided here are pivotal for developing improved storage solutions and predictive models for grain quality over time.

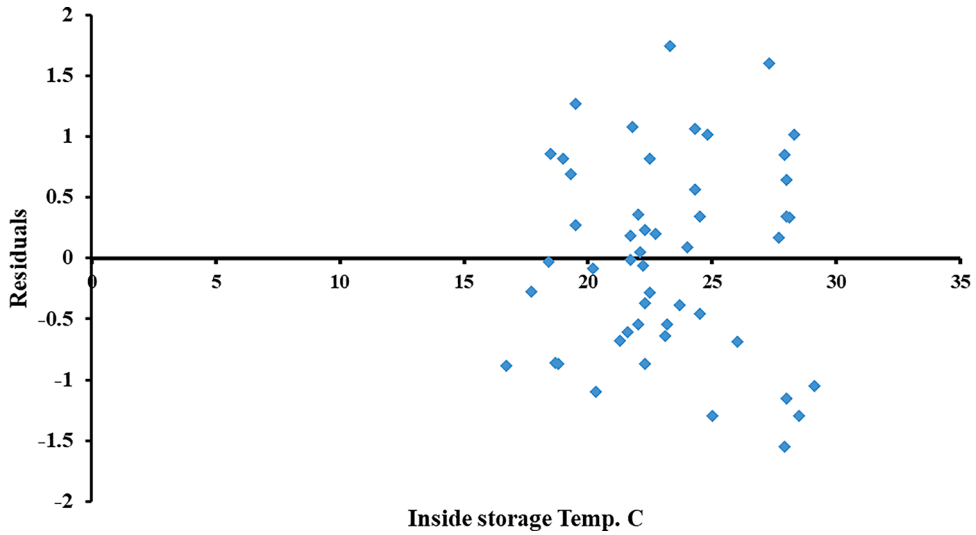


Fig. 9. Inside storage temperature residual plot.

4. Discussion

4.1. Temperature analysis

In-depth analysis of temperature dynamics within the GrainPro bag over a 125-day storage period reveals critical insights into the thermal behavior of the storage system. The fluctuation of room temperature from 16.1°C to 33.5°C, contrasted with the internal storage temperature range of 16.7°C to 29.2°C, underscores the bag's limited thermal insulation capabilities, attributed to its polyethylene composition. While the minimal temperature differential between the internal and external environments is notable, it also highlights a limitation in the material's ability to buffer against extreme external temperature variations. Future research could explore alternative materials or additional insulating layers to enhance the bag's thermal regulation properties.

4.2. Relative humidity insights

The study's examination of relative humidity patterns reveals a marked sensitivity to atmospheric moisture levels. The external RH's wide range (45 % to 81.2 %) compared to the more stable and insignificant variation of internal storage RH (45 % to 54 %) indicates the GrainPro bag's is effectiveness in humidity control which is not favorable condition to grow insect and other pest. However, this finding also underscores the need for more precise humidity regulation mechanisms, particularly in climates with extreme humidity variations.

4.3. Equilibrium moisture content (EMC) examination

Nuanced investigation into EMC dynamics highlights a gradual increase over the storage period, with regression analyses establishing a strong correlation with storage duration. This result is significant for understanding how storage time impacts grain moisture and quality. However, the study's limitation in not accounting for potential biochemical changes in the grain over time suggests an area for future research. Such studies could provide a more comprehensive understanding of the long-term effects of storage on grain quality.

4.4. Practical implications and predictive models

The derived regression models for internal storage EMC offer practical tools for grain storage management. The EMC model $(\text{EMC}\ (\%) = 12.1688 + 0.002547\ \text{Period}\ (\text{Day}))$ provides a predictive framework for anticipating moisture content changes. These models are invaluable for precision agriculture but should be validated under different climatic conditions and with various grain types to ensure their broader applicability.

4.5. Limitations

We acknowledge crucial limitations that surfaced during the research, particularly concerning the tools and methods used for environmental monitoring within the GrainPro bag. The digital hygrometer, a key instrument in the data collection process, presented significant challenges. It frequently delivered inconsistent and at times surprising data, especially in measuring relative humidity, a vital parameter in assessing storage conditions. This inconsistency suggests potential shortcomings in the sensor's accuracy. The device's inability to provide fractional humidity readings further limited the granularity of the data, impacting the precision of the analysis. Operational constraints also influenced the data collection. The necessity to conserve battery power led to irregular data recording intervals, potentially creating gaps in the dataset and affecting the continuity and completeness of the analysis. Additionally, poor lighting conditions inside the monitoring box hindered clear visibility of the hygrometer's readings, raising concerns about the accuracy of manually logged data. These limitations are essential considerations in interpreting study's findings. The study is also highlighted the need for more advanced and reliable monitoring tools in future research to enhance the accuracy and reliability of environmental data collection in grain storage settings.

4.6. Comparison with previous studies on hermetic storage technologies

Findings regarding the performance of the GrainPro bag in preserving grain quality and suppressing insect infestation are aligned with previous studies on hermetic storage technologies. Carvalho et al. (2019) conducted a comprehensive analysis of the effect of relative humidity on Sitophilus zeamais infestation in paddy rice stored under hermetic conditions. Their results demonstrated that hermetic storage systems, such as the GrainPro bag, create a modified atmosphere that suppresses insect populations and maintains rice quality by controlling relative humidity and temperature levels within the storage environment.

Similarly, Alam et al. (2022) evaluated the efficacy of hermetic GrainPro bags in comparison to traditional storage structures used by farmers in Bangladesh. Their study revealed a significant reduction in insect infestation and weight loss, as well as higher germination rates, in paddy seeds stored in GrainPro bags compared to traditional storage methods. Furthermore, Awal et al. (2018) conducted experiments to assess the performance of hermetic bags, including GrainPro, in storing rice over a five-month period. Their findings support the effectiveness of GrainPro bags in minimizing weight loss and insect infestation while preserving rice quality compared to traditional storage containers.

In the study, an innovative approach has been introduced to enhance the effectiveness of hermetic storage technologies, such as the GrainPro bag, through the implementation of a remote monitoring system. Traditional methods like gunny bags and plastic containers are common in Bangladesh for grain storage. However, recent advancements in hermetic storage technologies have significantly reduced postharvest losses of paddy. By integrating remote monitoring technology with hermetic storage solutions, the study contributes to sustainable agricultural practices by providing a practical, real-time solution to monitor and manage key environmental parameters. This advancement not only enhances the effectiveness of hermetic storage technologies but also ensures the preservation of grain quality over time, thus addressing the storage challenges faced by farmers and commercial actors worldwide. Incorporating remote monitoring technology allows for continuous monitoring and adjustment of storage conditions, optimizing the efficacy of hermetic storage solutions and further reducing postharvest losses.

5. Conclusion

This study successfully deployed an advanced monitoring system designed to accurately track real-time temperature and relative humidity fluctuations during the 125-day storage of paddy within the confines of a hermetic GrainPro bag. The data acquisition process, which occurred 50times, showcased the system's remarkable efficiency in providing a non-intrusive yet comprehensive understanding of the storage conditions. Notably, the approach eliminates the necessity of physically opening the GrainPro bag, underscoring its non-disruptive nature and practicality in real-world agricultural settings. The seamless transmission of data via WiFi further accentuates the technological prowess of the implemented monitoring system. This wireless connectivity not only enhances the ease of data retrieval but also establishes a foundation for future scalability and integration with broader agricultural technology networks. While acknowledging the inherent limitations of hygrometers in achieving pinpoint accuracy, the system emerges as an asset for elucidating the intricate dynamics of temperature and humidity within the storage environment. This nuanced comprehension of environmental variables becomes pivotal in assessing and mitigating potential risks to the quality of stored paddy, laying the groundwork for

informed decision-making in agricultural management. Crucially, the low-capital nature of the developed solution positions it as a pragmatic and accessible tool, particularly beneficial for resource-conscious Bangladeshi farmers. By empowering them with a cost-effective means of effectively managing challenges associated with insects and mold, the monitoring system not only addresses immediate concerns but also contributes to the long-term sustainability of grain storage practices in the region.

This study showcases the technical competence of the implemented monitoring system while underscoring its broader significance in tackling real-world challenges encountered by farmers. It adds to the expanding knowledge base in agricultural science and technology, presenting a scalable and practical solution with extensive implications for sustainable grain storage practices.

Furthermore, the technical analysis conducted in this study paves the way for future research, particularly in optimizing GrainPro bag storage for grain preservation. Key areas of interest include advanced humidity control methods and comprehensive assessments of grain quality over time, with a special focus on parameters such as germination viability and nutrient content. By addressing these aspects, not only refine the technical effectiveness of paddy storage systems, but also make significant contributions to the wider field of post-harvest grain management. This strategic approach not only overcomes the limitations of the current study but also establishes a roadmap for more detailed and practical research in grain storage technologies.

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Ethical statement for smart agricultural technology

Hereby, I, Md. Rostom Ali, consciously assures that for the manuscript 'Remotely Controlled Smart Monitoring System of Hermetic Paddy Storage to Reduce Postharvest Losses in Bangladesh' the following is fulfilled:

1) This material is the authors' own original work, which has not been previously published elsewhere.
2) The paper is not currently being considered for publication elsewhere.
3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
5) The results are appropriately placed in the context of prior and existing research.
6) All sources used are properly disclosed (correct citation). Literally copying of text is indicated as such by using quotation marks and giving proper reference.
7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.

I agree with the above statements and declare that this submission follows the policies of Smart Agricultural Technology as outlined in the Guide for Authors and in the Ethical Statement.

Date: 29–12–2023

CRediT authorship contribution statement

Md. Shariful Islam: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Sanjida Sadmani:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Md. Rostom Ali:** Supervision. **Nafis Sadique Sayem:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Md. Hamidul Islam:** Software, Formal analysis, Data curation. **Md. Abu Hanif:** Writing – review & editing, Formal analysis, Data curation. **Md. Monjurul Alam:** Writing – review & editing, Visualization, Supervision, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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